INCH-POUND
MIL-S-24340A(SH)
14 September 1988
SUPERSEDING
MIL-S-24340(SHIPS)
10 May 1968
(See 6.8)

MILITARY SPECIFICATION

SEALING COMPOUND, DECK, POLYURETHANE (POLYETHER BASE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a chemically-curing flexible material, containing a polyether polyurethane compound for embedment and seams of wood decks and underlayment, and seams of metal-clad deck panels.

1.2 Classification. The sealing compound shall be of the following types, as specified (see 6.2).

Type I - Compound having a fluid, self-leveling consistency.
Type II - Compound having a viscous consistency which is thixotropic in character.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet: General Specification for.
QQ-A-250/9	- Aluminum Alloy 5456, Plate and Sheet.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-704	- Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons).

MILITARY

MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.
FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-2073-1	- DoD Materiel Procedures For Development and Application of Packaging Requirements.

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations, Title 29, Part 1910.1200 - Hazard Communication Standard.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue

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of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1824 - Standard Test Method for Apparent Viscosity of Plastisols and Organosols at Low Shear Rates by Brookfield Viscometer. (DoD adopted)

D 2240 - Standard Test Method for Rubber Property-Durometer Hardness. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Type I sealing compounds furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample of type II sealing compounds shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material composition. Material composition shall be as follows:

(a) The base material shall be a liquid polyether type polyurethane-based polymer containing either isocyanate or mercaptan groups which, upon curing, shall result in a solid elastomeric compound. Sealing compound shall be a two-component system which requires a separate curing agent and mixing before curing. Sealing compounds shall cure without the addition of heat or pressure and shall not depend on solvent evaporation.

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(b) Priming materials shall be liquid that can be applied to the deck seams by spraying, brushing, or swabbing. This may involve the use of more than one material or more than a one-part material. The use of a primer is optional with the manufacturer, based on the characteristics of the specific substrate material. However, if a surface primer is recommended by the manufacturer, its field use by the end user shall be required.

3.3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.7.16). Questions pertinent to this effect shall be referred by the contracting activity to the Naval Medical Command (NAVMEDCOM) who will act as an advisor to the contracting activity.

3.4 Material safety data sheet (MSDS). The contracting activity shall be provided a MSDS at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazard Communication Standard. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence. FED-STD-313 shall be modified and supplemented accordingly. The MSDS shall be included with each shipment of the material covered by this specification (see 6.6).

3.5 Storage life. The components of the sealing compound as furnished by the contractor, shall be stable for at least 12 months from the time of delivery when stored unopened at a temperature of 50 to 90 degrees Fahrenheit (°F) (10 to 32 degrees Celsius (°C)). At the end of 12 months, freshly-extruded or mixed compound shall conform to the viscosity and application life requirements of table I, as applicable.

TABLE I. Physical and chemical requirements for sealing compound.

Property	Requirement	Test procedure
Specific gravity, maximum	1.50	4.7.3
Nonvolatile content, percent, minimum	97	4.7.4
Viscosity, poises, range: type I	200 - 700	4.7.5
type II	12,000 - 45,000	
Shrinkage, percent, maximum	3	4.7.6
Application rate, (type II only) grams per minute, minimum:	30	4.7.7
Curing rate, hours, maximum		
At standard conditions, (see 4.7.1)	48	4.7.8.1
At 40 ± 2°F	60	4.7.8.2
Hardness		4.7.9
Initial, range, durometer units	20 - 35	4.7.9.1
After oven aging, range, durometer units	20 - 45	4.7.9.2
Adhesion in tension		4.7.10.1
Initial, range, pounds per square inch (lb/in^2)	75 - 120	4.7.10.3
Ultimate elongation, minimum, percent	250	
Type of break, percent cohesion	85	

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TABLE I. Physical and chemical requirements for sealing compound. - Continued

Property	Requirement	Test procedure
After aging, range, lb/in^2	70 - 120	4.7.10.4
Ultimate elongation, percent, minimum	200	
After water immersion, range, lb/in^2	75 - 120	4.7.10.5
Ultimate elongation, percent, minimum	250	
Volume swell, after solvent immersion, percent, maximum	20	4.7.11
Blister size, inch, maximum	0.25	4.7.12.1
Recovery, percent, minimum	65	4.7.13
Low temperature resistance	No cracking, checking, or loss of adhesion	4.7.14

3.6 Physical and chemical requirements. The sealing compound shall conform to the requirements specified in table I.

3.7 Serviceability. Type I sealing compound shall perform satisfactorily for a minimum of 6 months when applied in wood deck seams aboard ship as specified in 4.7.15. The material shall not blister, separate from the wood or show any other signs of failure when examined during and after the shipboard service test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA), and shall consist of the tests specified in 4.7.

4.4 First article inspection. First article inspection shall consist of the tests specified in 4.7, except serviceability test (see 4.7.15).

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.2 Lot. For purposes of sampling, a lot shall consist of one production batch of base material, but shall not exceed 500 gallons.

4.5.3 Sampling procedure for quality conformance tests. For each lot of base material supplied under a contract or order, the contractor shall provide a kit or kits containing enough material to conduct all the quality conformance tests specified in 4.6.2. The kits shall consist of base material, and if part of the material system, an appropriate amount of curing agent and primer to use with the base material. The components of the kits shall be certified to have been taken from the components used in the lot of material offered for delivery.

4.5.4 Sampling for quality conformance inspection. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level S-1 and acceptable quality level (AQL) of 2.5 percent defective to verify conformance to all requirements regarding fill, closure, marking, and other requirements not requiring tests.

4.6 Inspection and tests.

4.6.1 Quality conformance inspection of containers. Each sample-filled container selected in accordance with 4.5.4 shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each filled container shall also be tested to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan specified in MIL-STD-105, the lot represented by the sample shall be rejected.

4.6.2 Quality conformance tests. The following tests shall be conducted on each kit provided in accordance with 4.5.3 as applicable:

Tests	Paragraph
Viscosity	4.7.5
Application rate	4.7.7
Curing rate	4.7.8

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Tests	Paragraph
Hardness (initial only)	4.7.9
Adhesion in tension (initial only)	4.7.10

4.6.2.1 Rejection. If a kit representing a lot is found not to be in conformance to this specification, the entire lot shall be rejected.

4.7 Test procedures.

4.7.1 Test conditions. Standard laboratory testing conditions shall be 73.5 ± 2°F (23 ± 1.1°C) and 50 ± 5 percent relative humidity. Except as otherwise specified hereinafter, all test specimens shall be prepared, cured, and tested under these conditions.

4.7.2 Preparation of test specimens.

4.7.2.1 Surface preparation. Surfaces on which the compound is to be applied shall be free of oil, grease, wax, or other contamination, and coated with the appropriate primer in accordance with the contractor's recommendation.

4.7.2.2 Preparation of compound. In the following test procedures, the term "fresh compound" refers to one-component material which has just been extruded from its cartridge or two-component materials that have just had the base material and curing agent thoroughly mixed together. For two-component material, all the base material and curing agent provided in the kit specified in 4.5.3 shall be mixed as thoroughly as possible. Care shall be taken in the mixing to have a minimum inclusion of air.

4.7.2.3 Curing conditions. Unless otherwise specified herein, specimens shall be cured for 72 ± 1/2 hours at standard test conditions (see 4.7.1) prior to testing.

4.7.3 Specific gravity. Three specimens of fresh compound molded into plugs approximately 1-11/16 inches in diameter by 3/4 inch thick shall be cured for 72 ± 1/2 hours. Care shall be taken to avoid air entrapment. The specimens shall be weighed accurately to three decimal places in grams after which they shall be dipped in methyl alcohol, and while wet, immediately suspended in distilled water at standard conditions and reweighed. The specific gravity shall be computed from the following formula:

$$\frac{\text{Weight in air}}{\text{Weight in air - weight in water}} = \text{Specific gravity}$$

4.7.4 Nonvolatile content. Five to 10 grams of fresh compound shall be transferred as rapidly as possible to a cup approximately 3 inches in diameter and 3/4 inch in depth. A fitted cover shall immediately be placed over the cup, and the weight determined using an enclosed beam balance which will weigh accurately to the nearest milligram. The weight of the cup and cover shall be determined prior to using, and subtracted from the initial and final weights in order to calculate the net sample weights. The cover shall then be removed, and the sealing compound heated in an oven for 24 hours at 158 ± 2°F (70 ± 1.1°C).

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It shall then be cooled in a desiccator, the cover replaced, and the weight determined. The percentage of total nonvolatile content shall be calculated as follows:

$$\text{Percent of nonvolatile content} = \frac{\text{Final weight}}{\text{Initial weight}} \times 100$$

4.7.5 Viscosity. One pint of fresh compound shall be tested immediately in accordance with ASTM D 1824, using a spindle no. 5 at 10 revolutions per minute (r/min) for type I material, and spindle no. 7 at 1 r/min for type II material.

4.7.6 Shrinkage. Fresh compound shall be placed in three containers approximately 1-11/16 inches in diameter by 3/4 inch deep, and filled level with the brim to obtain a definite volume of compound. After the compound has set for 96 ± 1/2 hours, the volume shall be determined by immersion of the filled cups in water. Shrinkage shall be expressed as the percentage of loss in volume with respect to the original volume of the compound. The original volume shall be taken as the internal volume of the cups.

4.7.7 Application rate (type II only).

4.7.7.1 Conditioning and mixing. The base material, curing agent, and application gun shall be stabilized at standard conditions (see 4.7.1) for at least 8 hours before a minimum of 250 grams of base material is mixed with a proper amount of curing agent. The mixed sealing compound shall be promptly used to fill the cartridge of the application gun, a standard Semco sealing compound gun, or equal, having a Semco 440 nozzle with an orifice diameter of 0.125 ± 0.0005 inch. The gun and sealing compound shall be maintained at standard conditions (see 4.7.1) throughout the test.

4.7.7.2 Test procedure. The gun shall be attached to a constant air supply of 90 ± 5 lb/in^2 gauge. Two to 3 inches of sealing compound shall be extruded initially to clear trapped air. When exactly 2 hours have elapsed from the time mixing of the components was started, the sealing compound shall be extruded onto a tared sheet of cellophane for exactly 1 minute using the specified gun pressure. The extruded sealing compound shall then be weighed and the result recorded as grams per minute of extruded compound.

4.7.8 Curing rate.

4.7.8.1 Standard temperature. Components and molds shall be preconditioned for at least 8 hours and tested under standard conditions (see 4.7.1). The time shall be noted when fresh compound is cast into a mold to provide a specimen 1/4 inch thick and at least 3/4 inch in diameter. The elapsed time to reach a minimum hardness of 10 durometer units shall be recorded. Hardness shall be determined on a double back-to-back specimen by the method specified in 4.7.9.

4.7.8.2 Low temperature. Components and container shall be preconditioned and tested at a temperature of 40 ± 2°F (4.4 ± 1.1°C). The time shall be noted when 40 to 50 grams of fresh compound is placed in a container approximately 2-1/4 inches in diameter. The compound shall be periodically probed full depth

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with a clean 1/4-inch diameter glass rod. The elapsed time shall be recorded when the compound releases completely from the probe within a withdrawal distance of 1 inch above the compound.

4.7.9 Hardness. Hardness shall be determined in accordance with ASTM D 2240.

4.7.9.1 Initial. Initial hardness shall be determined after the compound is allowed to cure for 96 ± 1/2 hours at standard conditions (see 4.7.1). The same specimen prepared for the test specified in 4.7.8.1 may be used.

4.7.9.2 Aged. The same specimen used in the test specified in 4.7.9.1 shall be immediately heat-aged for 7 days at 200 ± 2°F (93.3 ± 1.1°C). After this period, the specimen shall be allowed to cool at standard conditions (see 4.7.1) for not less than 16 hours and not more than 48 hours before determining hardness.

4.7.10 Adhesion in tension.

4.7.10.1 Specimens. Adhesion in tension shall be determined on nine different sets of specimens, three sets each for Douglas fir, teak, and aluminum. One set is required for initial determination, one for oven aging, and one for salt water immersion. Five specimens are required for each test set. Results for each specimen shall be recorded, but the five specimen values shall be averaged and this average value shall be used to determine if the set conforms to the applicable requirements specified in table I.

4.7.10.2 Preparation. The Douglas fir and teak test blocks shall be as shown on figure 1 and the wood shall be 15 to 20 percent moisture content. The two halves of the blocks shall be coated with primer in accordance with the contractor's recommendations. The aluminum test blocks shall be as shown on figure 2 and the aluminum alloy shall conform to QQ-A-250 and QQ-A-250/9. The aluminum shall be thoroughly cleaned with methyl ethyl ketone, allowed to dry at standard conditions (see 4.7.1) for 30 minutes, and then primed in accordance with the contractor's recommendations. A 1/4-inch layer of fresh compound shall be introduced between the two halves of each primed specimen and each specimen cured as specified in 4.7.2.3. The specimens shall then be further conditioned as specified in 4.7.10.3, 4.7.10.4 and 4.7.10.5 and subjected to a tensile load at a rate of 0.25 inch per minute.

4.7.10.3 Initial. One set of specimens prepared as specified in 4.7.10.2 shall be subjected to strain immediately after the cure period. For quality conformance tests (see 4.6.2) adhesion shall be determined on Douglas fir or aluminum specimens only, as specified by the acquisition activity (see 6.2).

4.7.10.4 Aged. One set of specimens prepared as specified in 4.7.10.2 shall be placed in a hot air circulating oven at 158 ± 2°F (70 ± 1.1°C) immediately after the cure period. It shall be kept in the oven for 166 ± 1 hours, then removed and subjected to strain within 5 minutes after removal from the oven.

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4.7.10.5 Immersed. One set of specimens prepared as specified in 4.7.10.2 shall be immersed for 96 ± 1/2 hours at standard conditions (see 4.7.1) in a 10 percent solution of salt. The specimens shall then be removed from the solution, rinsed with fresh water, wiped dry, and subjected to strain within 5 minutes after removal from the salt solution.

4.7.11 Swelling test. Two 1/8- by 1- by 2-inch specimens shall be tested in accordance with method 6211 of FED-STD-601. The immersion medium shall be fuel conforming to grade JP-5 of MIL-T-5624.

4.7.12 Blister resistance.

4.7.12.1 Preparation. The test arrangement, as shown on figure 3, shall consist of two pieces of Douglas fir at 15 to 20 percent moisture content embedded in the product in such a way that the bottom face and included seam are filled with the product. The freshly filled test specimen seam shall be exposed to a surface temperature of 150 ± 2°F (65.6 ± 1.1°C) by means of an infrared lamp located above the specimen. A 250-watt infrared bulb in a reflector located approximately 14 inches above the specimen may be used. After the specimen has been exposed for 6 ± 1/4 hours, the seam shall be examined for presence of blisters, bubbles, or porosity exceeding the size limitation specified in table I.

4.7.13 Recovery. Two test specimens as shown on figure 1 of Douglas fir of 15 to 20 percent moisture content shall be halved and coated with primer. A 1/4-inch layer of freshly mixed compound shall then be introduced between the halves and cured as specified in 4.7.2.3. Then they shall be subjected to a tensile load at a rate of 0.25 inch per minute until extended 150 percent. The specimen shall be held in this extended condition for 24 ± 1/4 hours, after which time it shall be removed and allowed to recover without compression by placing it edgewise on a glass surface with the long axis of the compound head perpendicular to the surface. Recovery shall be measured 60 ± 5 minutes after release from the stress and calculated by the following formula:

$$\frac{\text{Width extended - width after recovery}}{\text{Width extended - width initially}} \times 100 = \text{Recovery}$$

4.7.14 Low temperature resistance. Two test specimens each on Douglas fir, teak, and aluminum shall be prepared as specified in 4.7.10.2. After curing, material shall be extended to 100 percent of its original thickness at 20 ± 2°F (minus 28.9 ± 1.1°C) over a 4-hour period and held in the extended position for 16 hours at 20 ± 2°F (minus 28.9 ± 1.1°C). After the conditioning period, holding device shall be removed, and specimens shall be removed to standard conditions (see 4.7.1). Specimens shall be inspected for cracks, checking, and loss of adhesion.

4.7.15 Serviceability. The type I sealing compound shall be used on board ship to calk wood deck seams for a minimum period of 6 months. During and immediately after this service period, the calking shall be examined for blisters, separation from the wood, or any other signs of failure.

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4.7.16 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Naval Medical Command, MEDCOM 242, Washington, DC 20372 (see 3.3.1). The disclosure of proprietary information, which shall be held in confidence by the Naval Medical Command, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material. Information submitted shall be clearly marked or identified to show it is being provided in connection with qualification under MIL-S-24340.

4.8 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

5.1.1.2 Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flammability and smoke requirements specified in PPP-F-320.

5.1.2 Unit pack quantity. The compound shall be furnished in 1-quart, 1-gallon, or 5-quart cans, or 5-gallon pails as specified (see 6.2). The liquid curing agents shall be packaged in cans or glass containers in the quantity required for mixing the compound. The primer, when furnished, shall be packaged in cans or glass containers specified for curing agents.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Unit containers.

5.2.1.1.1 Cans. Cans shall conform to type V, class 2 or 4 of PPP-C-96. Exterior plan B coating and side seam striping is required. Cans of 1-gallon and 5-quart capacity shall be provided with handles which shall be galvanized or protectively coated to resist corrosion.

5.2.1.1.2 Pails. The 5-gallon pails shall conform to type II, class 3 of PPP-P-704. Wire handles and pails shall be zinc-coated or otherwise protectively coated to resist corrosion.

5.2.1.1.3 Glass containers. A commercial narrow-mouth glass container shall be used. The container shall be provided with a metal screw cap closure fitted with a nonabsorbent liner. A gasket which does not affect, and is not affected by, the packaged contents shall be provided to assure an airtight seal between the screw cap and the glass container.

5.2.1.1.3.1 Intermediate packs. Glass containers shall be intermediate packed in fiberboard boxes in accordance with PPP-B-636, class weather-resistant. Boxes shall have partitions, liners, full size top and bottom pads and separators constructed of the same material as used in the box. Separators shall be placed between all layers of unit packages. The cells formed by the partitions shall be snug-fitting and shall prevent movement of the glass container. If necessary, additional cushioning shall be used to prevent movement within the boxes. Closures of the boxes shall be in accordance with method V of the appendix of the box specification. Gross weight shall not exceed 20 pounds.

5.2.2 Level C. Preservation shall be as specified for level A, except that fiberboard boxes shall be of the non-weather-resistant domestic type with box closure in accordance with method I.

5.2.3 Commercial. Preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Cans. Cans shall be packed in accordance with the level A requirements of the appendix to PPP-C-96. Cans packed in tiers shall have a fiberboard pad in accordance with PPP-B-636, with compliance symbol W6 placed between the tiers.

5.3.1.2 Pails. Pails are not required to be overpacked. When specified (see 6.2), pails shall be palletized in accordance with MIL-STD-147.

5.3.1.3 Glass. Glass containers shall be packed in containers as specified in the exterior shipping container tables of MIL-STD-2073-1 for level A packing.

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5.3.2 Level B.

5.3.2.1 Cans. Cans shall be packed in accordance with the level B requirements of the appendix to PPP-C-96. Cans packed in tiers shall have a 200-pound minimum bursting strength. Class weather-resistant fiberboard pads in accordance with PPP-B-636 shall be placed between tiers.

5.3.2.2 Pails. Pails are not required to be overpacked. When specified (see 6.2), pails shall be palletized in accordance with MIL-STD-147.

5.3.2.3 Glass containers. Glass containers shall be packed in containers as specified in the exterior shipping container tables of MIL-STD-2073-1 for level B packing.

5.3.3 Level C.

5.3.3.1 Cans. Cans shall be packed in accordance with level C requirements of the appendix to PPP-C-96.

5.3.3.2 Pails. Pails are not required to be overpacked. When specified (see 6.2), pails shall be palletized in accordance with MIL-STD-147.

5.3.4 Commercial. Packing shall be in accordance with ASTM D 3951.

5.4 Marking. In addition to any special marking required (see 6.2), and such as specified in 29 CFR 1910, interior packs, exterior shipping containers, and palletized loads shall be marked in accordance with MIL-STD-129 and shall include applicable bar code markings. The date of manufacture and the batch or lot number shall be included in the marking information. Individual components shall be labeled to clearly identify one from another, and the largest volume component shall have a label or tag attached with the following instructions:

(a) Instructions for preparation of the compound, including correct proportioning of individual components.
(b) Instructions for application and proper use of the material, including any limitations on the storage temperature working life.
(c) Instructions for cleaning of equipment used for application.
(d) Any required safety precautions and procedures.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I sealing compound has a fluid, self-leveling consistency which is intended for use as follows:

(a) An embedment and seam sealant for wood decks installed over steel on aircraft carriers.
(b) A sealant for metal expansion seams, including underlayment and seams of metal clad deck panels.
(c) A gap filling sealant for general use where a solvent, heat- and weather-resistant elastic seal is required.

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6.1.2 Type II. Type II sealing compound is intended for use where a thixotropic, non-sagging characteristic is required, such as on inclined or vertical surfaces.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of sealing compound (see 1.2).
(c) Quantity of base materials required.
(d) When first article inspection is required (see 3.2).
(e) Material to be used for adhesion specimens (Douglas fir or aluminum) (see 4.7.10.3).
(f) When fire-retardant requirements are not required (see 5.1.1.1).
(g) Size and type container required (see 5.1.2).
(h) Level of preservation and packing required (see 5.2 and 5.3).
(i) If palletization is required (see 5.3.1.2, 5.3.2.2 and 5.3.3.2).
(j) Special marking required (see 5.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24340 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Material disposal. Disposal of the material components or the reacted material should comply with Federal, state, and local regulations.

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6.6 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

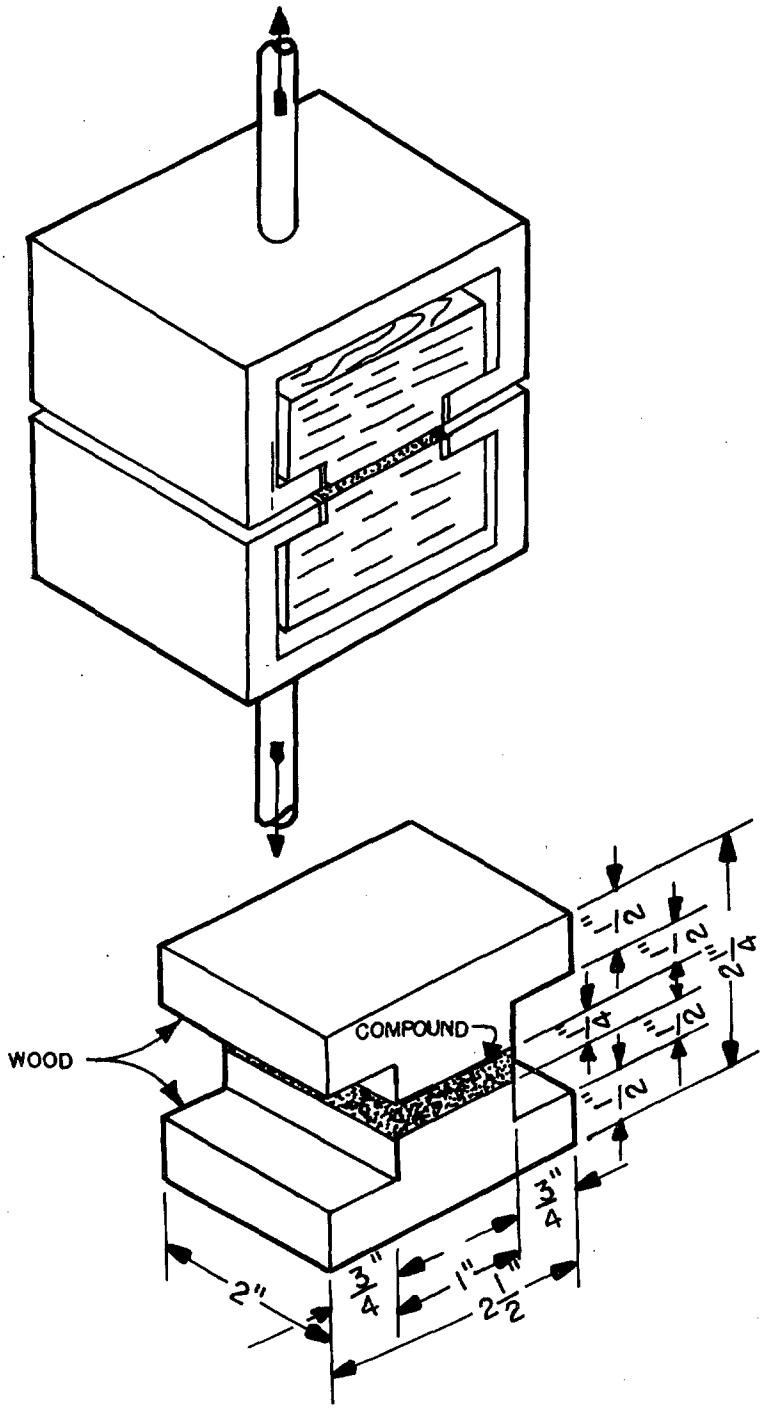
6.7 Subject term (key word) listing.

Elastomeric compound
Thixotropic
Viscosity

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8030-N097)

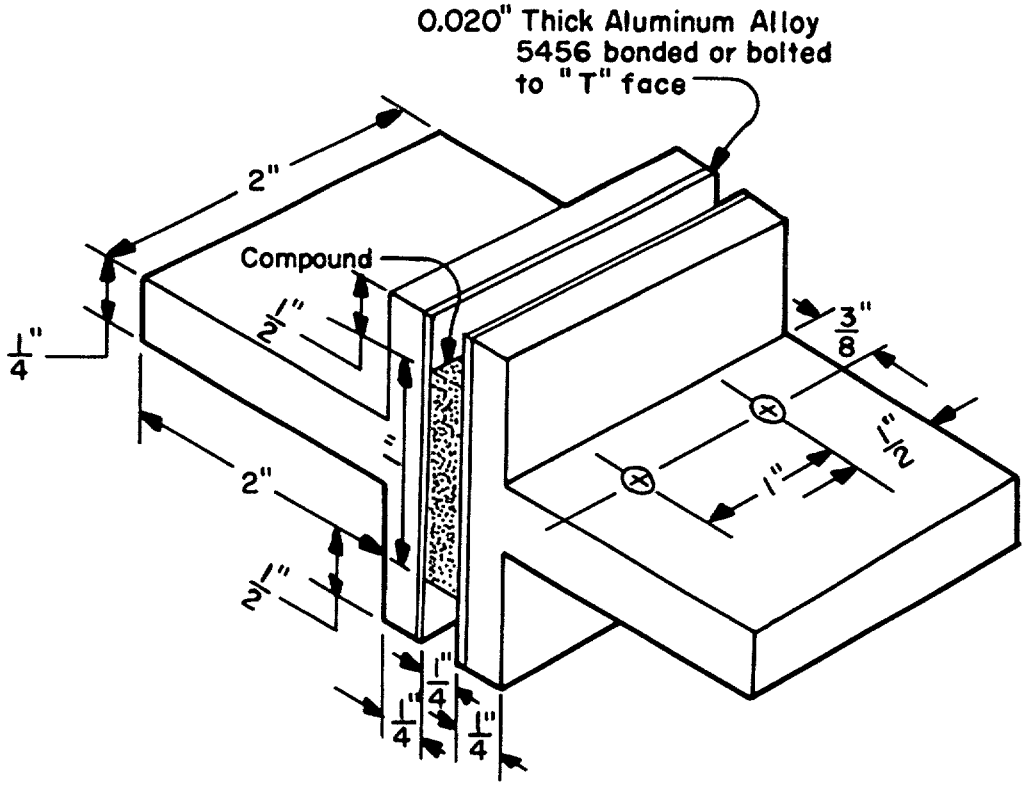
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SH 132031657

FIGURE 1. Diagrammatic sketch of method of conducting tension-perpendicular-to-grain test, with details of test specimen.

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SH 132031658

FIGURE 2. Assembly for testing adhesion in tension on aluminum.

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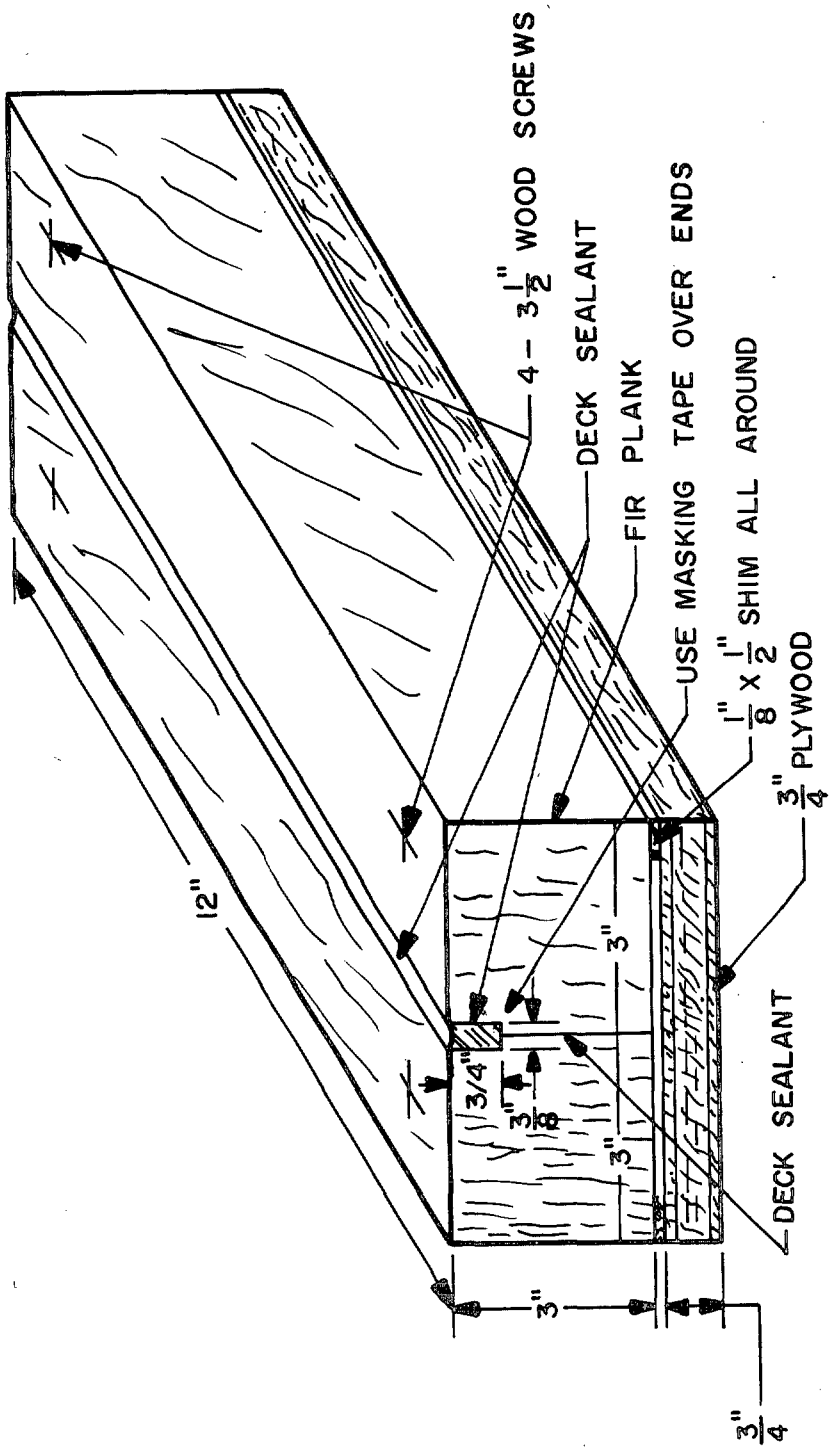


FIGURE 3. Blister resistance test specimen.

SH 132031659

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24340A(SH)	SEALING COMPOUND, DECK, POLYURETHANE (POLYETHER BASE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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